

Class 30006 – Financial Markets and Institutions
Università Commerciale Luigi Bocconi
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The Stock Market

(Chapter 13)



The Stock Market

- ✓ The **stock market** is the other capital market together with the bond market
- ✓ It receives a lot of media attention:
 - *“The market is down 2%”. Implicit: stock market*
- ✓ Why so much attention to stocks?
 - Because the gains are potentially infinite, and people love to get rich!
- ✓ Also, easy “see” the stock market:
 - floor traders shouting and big screens with lots of numbers to represent panic

Stock Market on the News



Source: La Repubblica, Aug 25th 2015, Chinese and global stock market plunge

Great Fortunes are made...

Market Summary > Tesla Inc

299,00 EUR

+ Follow

+196.02 (190.35%) ↑ past 5 years

11 Sept, 15:51 CEST • Disclaimer



\$1,000 worth of Tesla stocks bought 4 years ago
are worth \$3,000 today ...
... but also much more only Jan 20205!!! 🤪

...but also lost

Lehman Brothers Holdings Inc. (LEHMQ.PK)

3:59PM EDT: **0.0397** ↑ 0.0007 (1.79%)

Enter Symbol

GET CHART

COMPARE

EVENTS ▾

TECHNICAL INDICATORS ▾

CHART SETTINGS ▾

RESET

■ LEHMQ.PK



Chapter Preview

Topics of today

1. Stocks vs Bonds
2. Types of stocks
3. Where and how do stocks trade
4. Stock Market Indexes

Next class we will look at the valuation of stocks

Equity/Stocks/Shares

- ✓ A **share** of **stock** (or **equity**) represents a *fraction* of ownership in a firm:
 - a stock-/share-/equity holder owns a % of the firm
- ✓ Owner also **earns** ...
 - ... if there are profits to be shared (**dividends**)
 - ... if stock **price** increases
- ✓ Much **riskier** than bonds because:
 - **dividends** do not follow a fixed schedule or amount (like coupons do)
 - stock **prices** are more volatile than bonds, on average
- ✓ Ownership entitles also the stockholder to say something on firms' actions (*vote*)
- ✓ There is no maturity date!

Stocks vs Debt

- ✓ Stockholders are **residual claimants** ...
 - ... *i.e.* in a default they get what's left after all other creditors have been paid off
- ✓ Last time we discussed that bondholders themselves can then be subdivided into *senior* and *junior*
 - the order of payment is called **seniority** ...
 - ... **senior are paid before, junior after**
 - debt is senior to equity because it gets paid first
- ✓ ... that's the case for stockholders too (see next on common Vs preferred stocks)

2 Main types of Stocks

✓ Common Stock:

- a. *dividend*: holder is entitled to **discretionary** dividend payments
- b. *voting right*: holder **has** right to vote (... on the board of directors)
- c. *seniority*: it is a *residual claim* ...
 - ... holder is paid only after other claims (of bondholders and preferred stockholders have been met)

✓ Preferred Stock:

- a. *dividend*: pays a dividend, but it is **fixed** (like a bond)
 - thus, price is stable bcs easier to forecast future cash flows (see next lecture)
- b. *voting right*: **no** voting right (unless dividends have not been paid)
- c. *seniority*:
 - higher seniority than common stockholders
 - but still lower seniority than bondholders

2 Main types of Stocks

Memo tables (but not sufficient for the exam!)

feature	Stocks	
	Common	Preferred
Dividends	discretionary	fixed
Voting right	yes	no (usually)
Seniority ⁽¹⁾	lower	higher
Seniority ⁽²⁾	lower	lower

(1): between the 2 types of stocks

(2): with respect to bonds

Seniority scheme (updated):

- 1st senior **bond**holders
- 2nd junior **bond**holders
- 3rd stockholders with **preferred** stocks
- 4th stockholders with **common** stocks

Stocks Vs Bonds

Stocks and bonds have a balanced **trade-off** of control rights and cash flows

1. Stocks

- **high** control rights (vote)
- **uncertain** cash flow (volatile price, uncertain dividend)

2. Bonds

- **low** control rights (other than in bankruptcy)
- **certain** cash flow (less volatile, certain interest)

Stocks vs Bonds: Taxation

Equity funding is **more costly** than debt funding. Why?

- ✓ Because **interest** paid from bonds is **tax-deductible**, while **dividends** from stocks are **not**!
- ✓ Moreover, equity holders demand higher rate of return than bondholders as compensation for risk

Example: Suppose a firm makes \$20M of gross income.
Either pays \$5M in dividends or in interest

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Debt financed:

gross income	\$20,000,000
- interest paid	\$ 5,000,000
= taxable income	\$15,000,000
- tax 30%	\$ 4,500,000
= net income	\$10,500,000

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Example: Suppose a firm makes \$20M of gross income.
Either pays \$5M in dividends or in interest. Tax rate= 30%.

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= taxable income	\$15,000,000
- tax 30%	\$ 4,500,000
= net income	\$10,500,000

Financed with preferred stock:

gross income	\$20,000,000
- interest paid	\$ 0
= taxable income	\$20,000,000
- tax 30%	\$ 6,000,000
= net income	\$14,000,000
- dividends	\$ 5,000,000
=	\$ 9,000,000

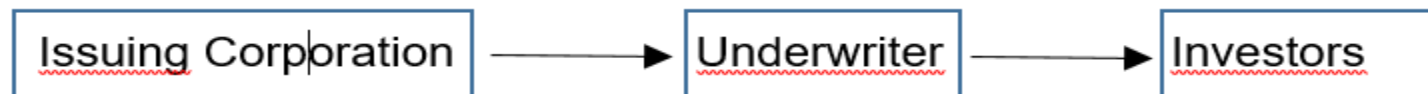
Shares Terminology

- ✓ **Authorized shares:** maximum number of shares the company can issue during its lifetime
 - decided at the time the firm files the registration statement
 - it can be changed, but it needs shareholders' vote
- ✓ **Outstanding shares:** number of shares company has issued
Outstanding shares include:
 - **restricted shares:** cannot be bought or sold by the public
 - these are the only type of stocks available in private companies (*i.e.* not yet listed)
 - usually given to employees and insiders as part of salaries or bonus
 - **'Float':** freely bought and sold without restrictions by the public
- ✓ Outstanding shares are typically used to calculate a company's

market capitalization = price per share × outstanding shares

Primary Market

- ✓ Def: new security issues sold to initial buyers
 - registration with Authority (SEC) is required before stock can be issued
- ✓ For issues of **new stock**
 - **Initial Public Offering (IPO)**: *first-time* offer of equity to be publicly traded
 - **Seasoned offering**: firm *already has* stocks trading in secondary market and issues new stocks
- ✓ Underwriter
 - Typically, it is an investment bank (syndicate if more inv. banks), that also arranges the sale of securities to investors
 - commitment underwriting: underwriter commits to buy full issue at fixed price from corporate issuer
 - best efforts underwriting: underwriter does not guarantee a price and acts more as a distribution agent



Secondary Market

- ✓ Securities previously issued are *bought and sold*
 - when people refer to “the stock market” they usually mean the secondary market for stocks
- ✓ Involves matching of buyers and sellers
 - it basically works like an auction market
- ✓ It can be an **organized exchange** (like the *NYSE*): a company that earn profit on trading fees and concessions
 - it used to imply a specific *physical* trading location, but **computer** systems have replaced the need for a physical location where traders meet (although they do still meet there)
 - main in the US: NYSE (best known with ≈ 4 bn shares daily), but also others (Chicago, etc ...)
 - Europe: London Stock Exchange, Deutsche Börse in Frankfurt, Bourse de Paris, Borsa Italiana in Milan etc..
- ✓ Or non-organized: an **over-the-counter (OTC)** market

Organized Exchanges

NYSE



Organized Exchanges

- ✓ Characterized as an **auction** market
 - physical trading floor with trading posts
- ✓ In the OE, each *trading post* has a **specialist**
 - each stock is assigned to one specialist
 - it acts as an **auctioneer** (sets the price) matching buyers and sellers
 - specialists are obliged to act as to stabilize order flow and prices
 - specialists belong to *specialist firms* and pay *concession fees* to NYSE
- ✓ **Floor traders/brokers** (representing clients) trade at the trade posts
 - usually with other brokers (intermediated by specialists) in 90% cases; only 10% directly with exchange specialist (takes ownership or sells from inventory)
 - **Market** orders: at market price, executed immediately
 - **Limit** orders: buy or sell a stock at a specific price or better

Trading posts

New York Stock
Exchange Trading Post,
1930



New York Stock
Exchange Trading Posts,
2024



Organized Exchanges

- ✓ To be listed at NYSE, a firm must satisfy *many* requirements
 - market capitalization > \$100 M (or \$40 M for an IPO)
 - number of shares outstanding > 1,1M
 - stock price $\geq$ 4\$
 - plus a number of further criteria (full list [here](#))
 - fees
- ✓ Being listed has some advantages:
 1. stocks trade on a *more liquid market* (can find easily many buyers and sellers)
 2. prestige
- ✓ Nevertheless, some big companies (Microsoft, Google!) never listed at NYSE ...
 - rather on NASDAQ Stock market in the US and as secondary listings other stock exch. in Europe (London, Frankfurt, Milan, etc...)

Over-the-counter markets (OTC)

- ✓ Lower listing requirements
- ✓ No trade floor, all electronic (NASDAQ is an OTC!)
- ✓ **Dealers** stand ready to “make a market” (*market makers*)
 - *many* dealers for each stock (competition)
 - they have inventory of stocks and quote two prices:
 - to **buy** at the **bid** price
 - to **sell** at the **ask** price
 - after entering bid and ask prices, dealers must trade a minimum quantity of shares at those prices
 - earn **bid-ask spread** and commission fees
- ✓ OTC important for thinly-traded securities (because need a market maker for trades to happen)

Stock Market Indexes

- ✓ Indexes capture the evolution of a group of shares
 - they are used to monitor the behavior of a groups of stocks.
 - “See how overall market is doing”
- ✓ Major **indexes** are:
 1. Dow Jones Industrial Average (*DJIA*): 30 companies
 2. the *S&P 500*: 500 companies
 3. the *NASDAQ composite*: over 3,000 companies
 4. *FTSE 100* (London): 100 companies
 5. Nikkei 225: 225 companies
 6. FTSE MIB (Milan): 40 firms
- ✓ These indexes are just averages (simple, value weighted) of the stocks composing the index
- ✓ The basket of underlying stocks changes periodically

Stock Market Indexes

- ✓ A **price weighted** index: *simple* arithmetic mean of prices

$$\frac{1}{N} \sum_{i=1}^N p_i$$

- every stock has the same weight
- ✓ The DJIA is an example of a price-weighted index, but it has adjustments. DJIA formula:

$$\sum_{i=1}^{30} p_i / (DowDivisor)$$

- a price weighted index corrected by *DowDivisor* (not based on market capitalization)
- the *DowDivisor*
 - keeps track of stock splits, dividends etc... to maintain historical continuity
 - at end of '24 was around 0.16 → multiplier $1/0.16=6.25$: big adjustment!

DJIA

- ✓ So, if the index is 46,200 it means that the average price (after adjustments) for one share in the DJIA is \$46,200
- ✓ In 1980 the DJIA was \$800: 57,3 times increase!
- ✓ Each stock in the Dow has the same weight, regardless of the company's market capitalization

Stock Market Indexes, S&P500

- ✓ S&P 500 is a value weighted index: weighted average of prices

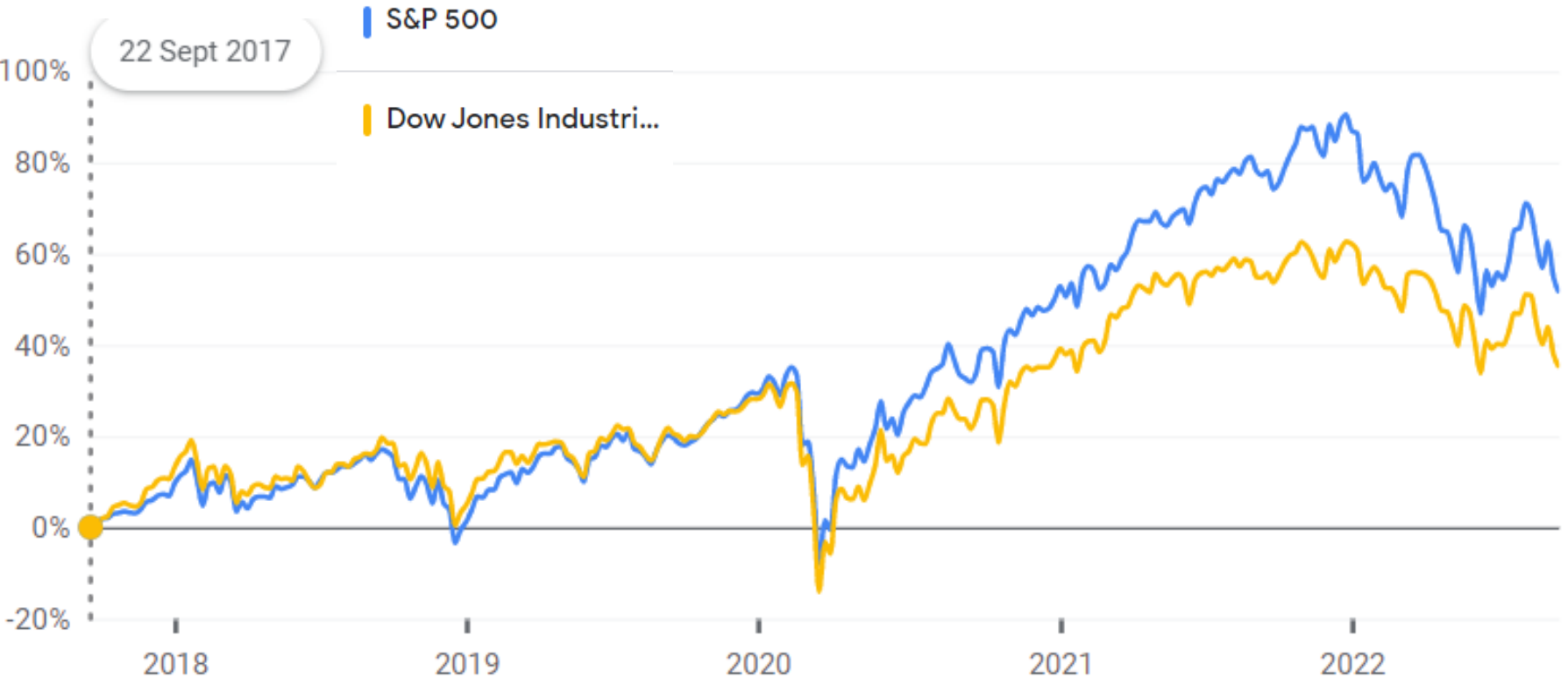
$$\frac{1}{mkt\ value\ base\ year} \sum_{i=1}^{500} (p_i \times n_shares_i)$$

- ✓ Every stock is weighted by its capitalization ($p \times q$, a composite index) over a base year
- ✓ So, if the index is **3,700**, it means ...
 - ... that the market value for firms in the S&P500 is **3,700** times higher compared to 1941-43 (base year)
 - in 1980 it was 100 – around 37 times increase
 - today S&P 500 is about 6,580

Co-movement of S&P500 and DJIA



Widening lately (because of tech stocks)



ETFs

Exchange-Traded Funds

- ✓ allow to **invest in stock market indexes**
- ✓ an investment fund that holds a basket of securities
 - automatically aligned with a stock market index they replicate (ex.: DJIA) ...
 - ... not actively managed
- ✓ very popular for investing in a diversified way at lower fees, also popular in foreign markets
- ✓ nicknames of ETFs tracing particular indexes. Fancy names:
 - Diamonds (DJIA)
 - Spiders (S&P 500)
 - Qubes (Nasdaq)